	School of Civil end Environmental Engineering	
	NUST Institute of Civil Engineering	
	Doc # MTS-CC-20 / **Date of Issue: 04-10-2012**	
	Lesson Plan	

SUBJECT: **ARCH-305: Architecture and Town Planning**

CREDIT HOURS: **2-0**

CONTACT HOURS: 2 Hours per Week

PREREQUISITE: NIL

MODE OF TEACHING: Lectures

COURSE DESCRIPTION: This course is designed to give the contemporary and ancient era concept of Architecture and Town Planning. Additionally, it concerns itself with research and analysis, strategic thinking on critical urban issues, architecture design and urban design, public participation, neighbourhood planning, understanding of urban planning process, Zoning, implementation and management.

COURSE OBJECTIVES: This course is meant to offer profound understanding of the forces that give meaning and values to the built environment through acts of design, construction and analysis. Moreover, it provides theoretical understanding of physical, social and economic aspects of human built environment, human settlements, land use planning and urban design. The course includes number of case studies from contemporary and ancient era.

RELEVANT PROGRAM LEARNING OUTCOME (S):

The course is designed so that students will achieve the PLO/s:

1	Engineering Knowledge:	☑	7	Environment and Sustainability:	☐
2	Problem Analysis:	☐	8	Ethics:	☐
3	Design/Development of Solutions:	☐	9	Individual and Team Work:	☐
4	Investigation:	☑	10	Communication:	☐
5	Modern Tool Usage:	☐	11	Project Management:	☐
6	The Engineer and Society:	☐	12	Lifelong Learning:	☐

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COURSE LEARNING OUTCOMES: Upon successful completion of the course, the student will be able to demonstrate competency by being able to:

1. Describe historical development of human settlement and distinguish different features and elements of Islamic, Greek and Roman architecture.
2. Understand concept of planning, objective of town planning, planning law, urban form and trends in urban growth with particular reference to Pakistan.
3. Understand suburban developments, urban renewal, planning process and planning surveys.
4. Classify Zoning and review the theories of land use patterns with particular reference to Pakistan.
5. Use of GIS in urban planning independently focusing on master planning preparation process.

PRACTICAL APPLICATION: Contemporary cities are confronted with numerous issues such as urban sprawl, emergence of katachi abadies/slums in the heart of the cities, poor service delivery etc. Contrary to this, cities are also the key component of national growth and considered as "engine for economic growth". This course provides insight understanding of the different issues pertaining to the urban areas and enables students to apply the various tools and techniques to solve these issues. Further, it helps the students to embed features of Islamic Architecture along with contemporary form by practical designing of neighbourhood units.

TOPICS COVERED:

Week	Sections Covered/Reading Assignment	Book reference
1	**Introduction to the course- Basic Planning Concepts** Definition, Aims, Scope, and Principles of Planning	Principles and Practice of Town and Country Planning by Lewis Keeble, pp 9-15
2	**Socioeconomic aspects of planning** Summary of elements of urban structure - Networks, Buildings, open spaces etc. The form of the modern city in the age of automobile - Inner city & Suburban area	Handouts
3	**Approaches to Planning** Overview about Industrial revolution, transition to the industrial city and paradigm shift to systems views of planning approach from comprehensive planning approaches	Handouts Regional Planning by john Glasson, pp 1-18
4	**Overview of Urban and Regional Planning Theories and Issues** Introduction to essential planning theories such as Location Theory, Concentric Zone Theory, Multiple-Nuclei Theory, Sector	Handouts

	Theory, Central Place Theory, Growth Pole Theory, etc.	
5	**Street patterns – different concepts** Layouts and location of various sectors in a city i.e., residential, commercial, gird iron pattern etc.	Handouts National Reference Manual on Planning and Infrastructure Standards by PEPAC Pakistan
6	**Street patterns – different concepts** Layouts and location of various sectors in a city i.e., residential, commercial, gird iron pattern etc.	Handouts National Reference Manual on Planning and Infrastructure Standards by PEPAC Pakistan
7	**The Preparation of the Plan (Master/structure Plan)** Description of various stages involve in physical planning process and the relation between surveys and plan.	Principles and Practice of Town and Country Planning by Lewis Keeble, pp 140-148
8	**Principles of planning and Planning Tools and Surveys** An introduction to basic planning principles and tools, i.e., cost-benefit analysis, planning balance sheet, goal achievement matrix, SWOT analysis, participatory approach and related surveys, i.e., land use survey, density survey, demographic surveys etc.	Handouts Principles and Practice of Town and Country Planning by Lewis Keeble, pp 113-126
9	**Current State, Challenges and Issues** General discussion on the current condition of cities and regions and the challenges and issues they are facing today in the form of Slums, Katchi abadies issues of inner city.	Handouts
10	**Urban renewal, Urban regeneration** Squatter settlement & slums up gradation, and re-planning the existing towns and cities	Handouts
11	**Planning Law and Legislation** Concept of Law, source of Law, meaning of norms of Law, legislation, ordinance bill, Act, President's consent, Regulations and Bye-Laws etc. Introduction of the roles of the various agencies as well as other stakeholders involved in the planning process.	Principles and Practice of Town and Country Planning by Lewis Keeble, pp 275-300 Handouts
12	**GIS application in Urban Planning** Describe how urban planners typically use GIS as a tool for analysis and the display of quantitative data to solve urban planning problems	Handouts
13	**Basics of Architecture** Pre-historic and Ancient East Architecture	A History of Architecture by Fletcher

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14	**Comparative study of** Egyptian, Greek, Roman, Gothic, and Indus Architecture & Civilization	A History of Architecture by Fletcher
15	Tracing the history and main **features of Islamic architecture**	A History of Architecture by Fletcher
16	**Study & Analysis of factors affecting architecture like** geographical climatic, politico religious, socio-cultural issues etc.	Handouts
17	**Development of buildings**, **use of materials** both for construction as well as finishing of surfaces e.g. stone, wood, metals, concrete, synthetic polymers & ceramics.	Handouts
18	**Revision Class/final exam**	

TEXT / REFERENCE BOOKS:

a. Broddbent. G. J.C. Anthony, Emerging Concept in Urban Space Design, Van Nostrand Reinhold Co. Ltd.
b. Snyter. J.C. Introduction to Urban Planning, Milwaukee McGraw Hill Book Co.
c. Fletcher. S.B. A History of Architecture, The Athlone Press
d. Lewis Keeble. Principles and Practice of Town and Country Planning
e. Kurt W. Bauer. City Planning for Engineers and Surveyors. CRC Press

GRADING POLICY:

Grades will assigned in the following manner:

Type of exam	Frequency	Weightage/test/exam/quiz	Percentage
One Hour test	2	15	30
Quiz	3	5	15
Assignments/project	3	5	15
Final Exam	1	40	40

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SCHEDULE: as posted

ROOM: Room No.114, NIT Building

OFFICE HOURS: Open Door Policy